

Why Rotating Fluids Make Polygons

Paper IV: The Havelock Field Theory

Gordon Speagle

Abstract

We construct a complete quantum field theory on the Seifert manifold $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$ with Thurston $\widetilde{\mathrm{SL}(2, \mathbb{R})}$ geometry, parameterised by a single integer N . The theory contains gravity, a $\mathrm{U}(1)$ gauge field, a charged scalar, and Dirac fermions, with all couplings determined by the central charge $c = N^2$: $G = 3/(2c)$, $\alpha = 6/(\pi c)$, $\alpha/(8\pi G) = 1/(2\pi^2)$ (exact, N -independent).

The cosmological constant $\Lambda = (N^2 - 16)/16$ is positive for $N \geq 5$, with $\Lambda = 0$ at $N = 4$ (the gauge threshold where $j = 1$). The Standard Model gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ emerges from the polygon hierarchy: $\mathrm{SU}(3)$ from the Galois cover at $N = 7$ ($\mathrm{Gal}(K_7/\mathbb{Q}) = \mathbb{Z}/3\mathbb{Z} = Z(\mathrm{SU}(3))$); $\mathrm{SU}(2)$ from the Chern–Simons adjoint Wilson line at $N = 4$ ($j = 1$, the adjoint of $\mathrm{SL}(2)$); $\mathrm{U}(1)$ from Kaluza–Klein reduction on S^1 . The Weinberg angle $\sin^2 \theta_W = 3/11 \approx 0.273$ (18% from the experimental value 0.231, versus the $\mathrm{SU}(5)$ GUT prediction $3/8 = 0.375$, which is 62% away).

The Weyl anomaly $\delta_m = 0$ exactly on all constant-curvature surfaces; the three-layer decomposition is exact with zero anomaly. The theory is UV-complete through the orbifold CFT at $c = N^2$ (non-perturbative) and super-renormalisable after Kaluza–Klein reduction to two dimensions.

1 Introduction: one integer, one geometry

The preceding papers established that the Havelock eigenvalue $\lambda_m = C_1(\xi) - m(N - m)/2$ governs both point vortex polygon stability (Paper II) and gravitational polygon stability in 2+1D (Paper III), with the palindromic thresholds encoding algebraic number fields (Paper I). Paper III derived the vacuum Einstein equations from polygon stability and identified the graviton at $N = 7$ ($j = 2$, the unique polygon number with integer spin ≥ 2 at the critical mode).

This paper constructs the full quantum field theory. The single input is an integer $N \geq 3$ (the polygon number = the flux quantum = the superselection sector). Everything else follows:

1. §2: the spacetime $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$ with Thurston $\widetilde{\mathrm{SL}(2, \mathbb{R})}$ geometry;
2. §3: gravity, gauge, scalar, fermion, radion;
3. §4: G , α , g_4 —all from $c = N^2$;
4. §5: the cosmological constant $\Lambda = (N^2 - 16)/16$;
5. §6: the Weyl anomaly vanishes ($\delta_m = 0$ on constant-curvature backgrounds);

6. §7: the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$;
7. §8: the Weinberg angle $\sin^2 \theta_W = 3/11$;
8. §9: the two-force unification (gravity/colour + electroweak);
9. §10: UV completion via the orbifold CFT.

2 The spacetime: $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$

The four-dimensional spacetime is the product of a time axis with a Seifert-fibred three-manifold:

$$ds^2 = -dt^2 + ds_{\mathbf{H}^2}^2 + (d\varphi + A_i dx^i)^2, \quad (2.1)$$

where $ds_{\mathbf{H}^2}^2 = d\rho^2 + \sinh^2 \rho d\theta^2$ ($K = -1$) and A is a $U(1)$ connection on the S^1 fiber with flux

$$\frac{1}{2\pi} \oint dA = \frac{N}{2}. \quad (2.2)$$

This is a Thurston geometry of type $\widetilde{SL(2, \mathbb{R})}$ [?]. The Kaluza–Klein reduction on S^1 decomposes any field into N modes with masses

$$\mu_m = |m - N/2|, \quad m = 0, 1, \dots, N-1. \quad (2.3)$$

The critical mode $m = N/2$ (even N) is massless: it IS the Havelock critical mode, and the Casimir $f(m, N) = m(N-m)/2$ IS the kinetic energy on S^1 .

3 The field content

The metric (2.1) yields, after KK reduction:

1. **Gravity.** The 2+1D metric g_{ij} on $\mathbf{H}^2$ is a Chern–Simons gauge field at level $k = c/12 = N^2/12$ (Proposition 2.1, Paper III).
2. **$U(1)$ gauge field.** The off-diagonal component A_μ from the KK ansatz; coupling $\alpha = 6/(\pi c)$.
3. **Charged scalar.** N KK modes Φ_m with masses $\mu_m = |m - N/2|$. The critical mode ($m = N/2$) is massless.
4. **Dirac fermion.** N KK modes Ψ_m with masses $\mu_m^f = |m - (N-1)/2|$ (shifted by $1/2$ from the spin connection). Even N : bosons have a massless mode, fermions do not $\implies$ SUSY broken by flux quantisation.
5. **Radion.** The S^1 modulus σ (fiber radius), stabilised by the flux energy at $\sigma = 1$.

4 The couplings: all from $c = N^2$

Proposition 4.1 (Exact coupling relations). *All couplings are determined by $c = 12b(N) = N^2 + O(N)$:*

$$G = \frac{3}{2c}, \quad (4.1)$$

$$\alpha = \frac{6}{\pi c}, \quad (4.2)$$

$$\frac{\alpha}{8\pi G} = \frac{1}{2\pi^2} \quad (\text{exact, } N\text{-independent}). \quad (4.3)$$

The quartic vertex from the Magri hierarchy: $g_4 = 2Nf^2/\pi$ at the critical mode.

Proof. G : from Brown–Henneaux, $c = 3\ell/(2G)$ with $\ell = 1$. α : from the KK gauge coupling on the unit-radius fiber. The ratio (4.3): $\alpha/(8\pi G) = [6/(\pi c)]/[8\pi \cdot 3/(2c)] = 6/(12\pi^2) = 1/(2\pi^2)$. The N -dependence cancels algebraically. $\square$

5 The cosmological constant

Proposition 5.1 (Cosmological constant from the fiber). *The effective two-dimensional cosmological constant is*

$$\Lambda = \frac{N^2 - 16}{16}. \quad (5.1)$$

$\Lambda = 0$ at $N = 4$ (the gauge threshold, $j = 1$); $\Lambda < 0$ for $N = 3$ (AdS); $\Lambda > 0$ for $N \geq 5$ (de Sitter).

Proof. The fiber curvature from the magnetic flux: $R_{\varphi\varphi} = (N/2)^2/4 = N^2/16$. The base curvature: $R_{\mathbf{H}^2} = -1$. The effective cosmological constant: $\Lambda = R_{\text{fiber}} - |R_{\text{base}}| = N^2/16 - 1 = (N^2 - 16)/16$. $\square$

The two thresholds have a unified interpretation:

N	Λ	j_{crit}	Physical meaning
4	0	$j = 1$	Gauge boson; $\Lambda = 0$ boundary
7	$33/16$	$j = 2$	Graviton; stability boundary $N_{\text{crit}} = 7$

6 The Weyl anomaly vanishes

Theorem 6.1 (Zero anomaly on Einstein backgrounds). *On any surface of constant curvature (the plane, $\mathbf{S}^2$, $\mathbf{H}^2$, or any quotient $\Gamma \backslash \mathbf{H}^2$), the constrained Havelock eigenvalue satisfies*

$$\lambda_m = C_1(\text{surface}, R) - \frac{m(N-m)}{2}$$

with $\delta_m = 0$ identically. The three-layer decomposition is exact with zero Weyl anomaly.

Proof. The Havelock identity (Paper I, Theorem 4.2) proves that the constrained eigenvalue (on $\{J = \text{const}\}$) has a mode-independent confining potential C_1 . The angular Hessian involves the csc^2 kernel from the logarithmic singularity of the Green's function; the Havelock sum (Paper I, Lemma 4.1) converts this to $m(N-m)/2$. No residual

δ_m survives. The key: on any constant-curvature surface, the Laplacian Green's function depends only on geodesic distance, so the mode decomposition is exact. Quotients $\Gamma \backslash \mathbf{H}^2$ add a mode-independent correction δC_1 (from the Selberg trace formula, Paper I, Theorem 7.1), which shifts C_1 but does not introduce mode-dependent anomalies. $\square$

7 The Standard Model gauge group

The polygon hierarchy realises $SU(3) \times SU(2) \times U(1)$ through three independent mechanisms.

7.1 $SU(3)$ from the Galois cover at $N = 7$

The trace field $K_7 = \mathbb{Q}(\cos 2\pi/7)$ has degree $[K_7 : \mathbb{Q}] = 3$. The Galois group $\text{Gal}(K_7/\mathbb{Q}) \cong \mathbb{Z}/3\mathbb{Z}$ acts on the three palindromic mode pairs of the $N = 7$ polygon:

Pair	Modes	Casimir f	Galois orbit
1	$(m=1, m=6)$	3	$1 \rightarrow 2 \rightarrow 3 \rightarrow 1$
2	$(m=2, m=5)$	5	$2 \rightarrow 3 \rightarrow 1 \rightarrow 2$
3	$(m=3, m=4)$	6	$3 \rightarrow 1 \rightarrow 2 \rightarrow 3$

The cyclic permutation identifies $\mathbb{Z}/3\mathbb{Z} = Z(SU(3))$ (the centre of $SU(3)$), with the three pairs as the fundamental representation. The full $SU(3)$ acts as unitary rotations on the three-dimensional pair space; the Casimir spectrum $(3, 5, 6)$ breaks $SU(3) \rightarrow U(1)^2$ (the Cartan subgroup).

Gravity = colour: on the base $\mathbf{H}^2$, the theory is gravity (the graviton $j = 2$). On the Galois cover $(\mathbf{H}^2)^3/\mathbb{Z}_3$, the same theory has $SU(3)$ gauge symmetry: three copies of $\mathbf{H}^2$, one per real embedding of K_7 , permuted by the Galois group. The Galois correspondence IS the gauge-gravity duality.

7.2 $SU(2)$ from the Chern–Simons adjoint at $N = 4$

Proposition 7.1 ($SU(2)$ from the adjoint Wilson line). *At $N = 4$, $m^* = 2$: $f(2, 4) = 2 = j(j+1)$ with $j = 1$. The $j = 1$ representation of $SL(2, \mathbb{R})$ is the adjoint (dimension 3). Wick rotation to the compact form gives $SU(2)$; the adjoint Wilson line carries $SU(2)$ gauge symmetry.*

7.3 $U(1)$ from Kaluza–Klein

The S^1 fiber with flux $N/2$ gives a $U(1)$ gauge field at all N , with coupling $\alpha = 6/(\pi c)$ and the exact lock $\alpha/(8\pi G) = 1/(2\pi^2)$.

8 The Weinberg angle

Proposition 8.1 (Weinberg angle from the polygon hierarchy). *At $N = 4$, the critical mode $m^* = 2$ carries KK hypercharge $Y = m^*/N = 1/2$ and normalised $SU(2)$ Casimir $C_2/(N-1) = f(2, 4)/3 = 2/3$. The Weinberg angle is*

$$\sin^2 \theta_W = \frac{Y^2}{Y^2 + C_2/(N-1)} = \frac{1/4}{1/4 + 2/3} = \frac{3}{11} \approx 0.2727. \quad (8.1)$$

The experimental value at M_Z : $\sin^2 \theta_W = 0.2312$. The SU(5) GUT prediction: $3/8 = 0.375$ (at the GUT scale). Our prediction $3/11 = 0.273$ is 18% above experiment, versus SU(5)'s 62%—substantially closer, and the remaining discrepancy is expected from renormalisation-group running from the polygon scale to M_Z .

9 The two-force unification

The four forces of the Standard Model reduce to two:

Force 1: Gravity/Colour ($N = 7$ sector). On the base $\mathbf{H}^2$: gravity (spin-2 graviton, coupling $G = 3/(2c)$). On the Galois cover $(\mathbf{H}^2)^3/\mathbb{Z}_3$: SU(3) colour ($g_s^2 = 3g_{\text{CS}}^2$, three colours from three Galois conjugates). Long range: gravity (curvature). Short range: colour (confinement = compactification of the cover). The Galois correspondence $K_7/\mathbb{Q}$ IS the gauge–gravity duality for this sector.

Force 2: Electroweak ($N = 4$ sector + **KK).** SU(2): from the CS adjoint Wilson line at $j = 1$ ($N = 4$). U(1): from the S^1 KK photon. Unified by the Seifert structure; the Weinberg angle $\sin^2 \theta_W = 3/11$ is determined by the KK hypercharge and the CS Casimir.

Unification field. Both sectors live in the cyclotomic field $\mathbb{Q}(\zeta_{56}) = \mathbb{Q}(\zeta_7, \zeta_8)$, which contains the trace fields $K_7 = \mathbb{Q}(\cos 2\pi/7)$ (gravity/colour) and $K_4 = \mathbb{Q}$ (electroweak) as subfields.

10 UV completion

The theory is UV-complete in the following precise sense.

Non-perturbative definition. The orbifold CFT at $c = N^2$ —a $\mathbb{Z}_N$ orbifold of N^2 free bosons on S^1 —is an exactly solvable 2D conformal field theory with known partition function, OPE coefficients, and conformal blocks. In the $\text{AdS}_3/\text{CFT}_2$ framework, this boundary CFT *defines* the bulk Chern–Simons theory at $k = N^2/12$ non-perturbatively.

Super-renormalisability. After KK reduction to 1+1D, the superficial degree of divergence is $D = 2 - E$ (E = external legs). Only the 2-point function ($D = 0$) has a logarithmic divergence; all vertices with $E \geq 4$ are UV-finite. Two renormalisation parameters suffice: the vortex core size a (UV cutoff for self-energy) and the central charge c (overall energy scale).

Physical observables. All physical observables—Havelock eigenvalues, partition function, couplings, cosmological constant, Weinberg angle—are *exact* (no perturbative approximation). The only divergences (self-energy at $d = 0$, KK zero-point sum) are renormalised by physical parameters within the framework.

11 Discussion

The Havelock Field Theory is specified by a single integer N on a single geometry $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$. It produces:

- Gravity + $SU(3) \times SU(2) \times U(1)$ gauge symmetry;
- $\Lambda = (N^2 - 16)/16$ ($\Lambda = 0$ at $N = 4$, positive for $N \geq 5$);
- $\sin^2 \theta_W = 3/11$ (18% from experiment);
- Exact coupling lock $\alpha/(8\pi G) = 1/(2\pi^2)$;
- Zero Weyl anomaly on Einstein backgrounds;
- UV completion via orbifold CFT;
- The graviton at $N = 7$ (unique);
- Two forces: gravity/colour + electroweak.

What it does *not* produce: the fermion mass hierarchy, the CKM matrix, the number of generations from first principles (though $\text{ind}(D) = N/2 + (1-g) = 3$ for $(N, g) = (4, 0)$, $(6, 1)$, or $(8, 2)$), or a resolution of the landscape (N is an input, not derived).

The theory is built from four classical ingredients: Havelock (1931), Kaluza–Klein (1921), Thurston (1982), and Chern–Simons (1989). No string theory. No extra dimensions beyond S^1 .

References